

Our LST code: Validation and Verification

An agreement audit of the solver against published benchmarks
at their exact stated conditions

Our LST code technical documentation

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1 Summary of all cases

Table 1 lists every case presented below, together with the solver route that produced the compared quantity: the formulation (temporal or spatial), the numerical route (collocation eigenvalue solve or shooting), and its input and output. The routes themselves are specified in the companion numerical-methods document; the caveats live in each subsection. The Malik temporal test cases 3, 4, and 5, previously logged non-reproducible for want of their dimensional conditions, are now included, once Malik’s own Table I supplied the missing T_0 and wall temperatures.

Table 1: Cases presented in this document, with the solver route behind each comparison. GEVP: the temporal generalised eigenvalue problem $A\phi = cB\phi$ (input a real wavenumber, output the complex spectrum); QEP: the spatial quadratic eigenvalue problem in α (input a real frequency, output the complex wavenumber); shooting: the exact first-order march with $\sigma_{\min} \rightarrow 0$ root search.

Case	Source	Quantity	Solver route (input $\rightarrow$ output)
<i>Incompressible core</i>			
Plane Poiseuille	Orszag [1]	O-S eigenvalue c	temporal O-S GEVP, QZ ($\alpha, Re \rightarrow$ least-stable c)
OS spectrum (32 modes)	Orszag [1], Tab. 5	full temporal spectrum	temporal O-S GEVP, QZ ($\alpha, Re \rightarrow$ full c spectrum)
<i>Compressible eigenvalue anchors</i>			
Malik case 6	Malik [2], Tab. IX	spatial α (2nd mode)	spatial QEP, shift-invert ($\omega \rightarrow \alpha$)
Malik cases 3/4/5	Malik [2], Tab. IV/V/VI	temporal ω (M2.5/10)	temporal GEVP, 3-D for the oblique case 3 ($\alpha, \beta \rightarrow \omega$)
Malik Table X	Malik [2], Tab. X	spatial α (M10 2nd mode)	spatial QEP, shift-invert ($\omega \rightarrow \alpha$)
Malik Fig. 4 eigenfn.	Malik [2], Fig. 4	M10 2nd-mode $\hat{T}(y)$ shape	temporal GEVP, eigenvector ($\alpha \rightarrow \hat{q}(y)$)
B&M branches	Balakumar-Malik [3]	discrete/continuous topology	spatial QEP, full companion spectrum ($\omega \rightarrow$ all α)
B&M via Xi-Ren-Fu	Balakumar-Malik [3]	spatial α (2nd mode)	spatial QEP ($\omega \rightarrow \alpha$)
<i>Second mode: curves & families</i>			
Mack Fig. 10.6	Mack [4]	$\omega_{i,\max}(R)$, M4.5–10	temporal GEVP swept in α per R ($\alpha \rightarrow \omega_i$; max over α)
Egorov M6	Egorov et al. [6]	most-amplified ω	spatial QEP swept in ω ($\omega \rightarrow \sigma = -\alpha_i$)
Ma & Zhong M4.5	Ma & Zhong [7]	neutral curve, spatial 2nd mode	spatial QEP at fixed F ($\omega \rightarrow \alpha_i$; $\sigma = 0$ crossings)
<i>First mode: curves & families</i>			
Mack Fig. 10.1	Mack [4]	neutral loop, M1.6	temporal GEVP on an (R, α) grid ($\alpha \rightarrow c_i$; $c_i = 0$ contour)
Mack Fig. 10.3	Mack [4]	$\omega_{i,\max}(R)$, M1.3/3.0	exact oblique shooting, continuation from an anchor root ($\alpha, \beta \rightarrow c$ via $\sigma_{\min} \rightarrow 0$)
Mack Fig. 10.4	Mack [4]	oblique 1st mode, M4.5–10	temporal 3-D GEVP ($\alpha, \psi \rightarrow \omega_i$; max over α, ψ)
<i>Mixed first + second mode</i>			
Özgen Fig. 3	Özgen-Kırçalı [5]	neutral-curve family (both modes), M2–10	temperature-form temporal GEVP + branch continuation ($\alpha_L \rightarrow c_i$; $c_i = 0$ contour)

2 Incompressible core

2.1 Orszag (1971): plane-Poiseuille Orr–Sommerfeld eigenvalue

Reference. Orszag [1], the canonical plane-Poiseuille temporal eigenvalue.

Conditions. Plane Poiseuille flow $U = 1 - y^2$ on $[-1, 1]$, Reynolds number $Re = 10\,000$, stream-wise wavenumber $\alpha = 1$, temporal formulation.

Solving the temporal Orr–Sommerfeld problem by Chebyshev collocation and dense QZ (input the real wavenumber α and Re , output the complex phase-speed spectrum), Our LST code reproduces the benchmark least-stable eigenvalue (phase speed) $c = 0.23753 + 0.00374i$ to an absolute error below 10^{-4} at $N = 64$, tightening toward $\sim 10^{-7}$ under refinement (Table 2); this is a scalar comparison against an essentially exact reference, so no overlay figure is needed. The same solver also recovers the Blasius Tollmien–Schlichting mode at $Re_{\delta^*} = 998$, $\alpha = 0.179$ ($c_r \approx 0.36$, unstable) and finds unstable growth at $Re = 1000$, confirming the mapped semi-infinite domain and the growth-rate optimisation loop.

Table 2: Plane Poiseuille temporal eigenvalue, $Re = 10\,000$, $\alpha = 1$. Target $c = 0.23753 + 0.00374i$ [1].

N	Our LST code $c = c_r + ic_i$	$ c - c_{\text{target}} $
64	$0.23753 + 0.00374i$	$\lesssim 10^{-4}$
96	$0.23753 + 0.00374i$	$\lesssim 10^{-5}$
128	$0.23753 + 0.00374i$	$\lesssim 10^{-7}$

2.2 Orszag (1971) Table 5: the full Orr–Sommerfeld spectrum

Reference. Orszag [1], Table 5: the 32 least-stable Orr–Sommerfeld eigenvalues of plane Poiseuille flow at $\alpha = 1$, $R = 10\,000$, each classified symmetric (S) or antisymmetric (A).

The same temporal Orr–Sommerfeld route as Section 2.1 is exercised on its full-spectrum output, returning *all* eigenvalues of the collocated pencil at once rather than just the single least-stable mode. Our LST code’s dense eigensolver reproduces *all* 32 tabulated eigenvalues to a maximum absolute error 7×10^{-6} (median 5×10^{-8}) at $N = 128$. It resolves the A-branch ($c_r \sim 0.96$ downward), the P-branch ($c_r \sim 0.2$ – 0.5), and the vertical S-branch, including the near-degenerate S/A pairs that differ only in the fifth–sixth digit (e.g. modes 2 and 3 at 0.96463 vs. 0.96464), exercising the dense full-spectrum path (Figure 1).

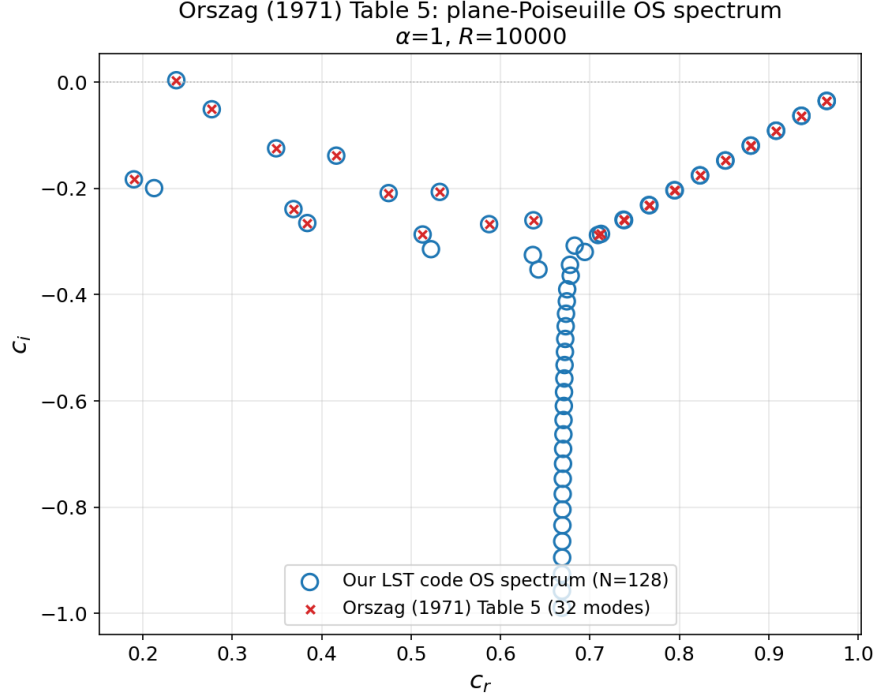


Figure 1: Orszag Table 5 [1]: Our LST code’s plane-Poiseuille Orr–Sommerfeld spectrum (circles) over the 32 tabulated eigenvalues (crosses), $\alpha = 1$, $R = 10000$. The classic “Y”-shaped spectrum (A-, P-, and S-branches), with every tabulated mode matched to $< 10^{-5}$.

3 Compressible eigenvalue anchors

3.1 Malik (1990): compressible eigenvalue anchors

Reference. Malik [2], the standard tabulated compressible-LST test-case suite. Case 6 (Table IX, the $M = 4.5$ spatial second mode) is the primary anchor, cross-checked digit-for-digit against Xi, Ren & Fu [8] and Hildebrand et al. [9]. Cases 3, 4, 5 (Tables IV/V/VI) and the grid-refinement case of Table X were long non-reproducible for want of their dimensional conditions ([9] lists only M and R), until Malik’s own Table I supplied the missing T_0 and wall temperatures, in particular case 4’s *cooled* wall $T_w/T_{adb} = 0.1$ that earlier attempts had assumed adiabatic.

Conditions. All cases share $Pr = 0.7$, $\gamma = 1.4$, Sutherland $S = 110.33$ K, L^* length scale. The suite spans $M = 2.5$ –10, temporal (fixed real α , complex ω) and spatial (fixed real ω , complex α) formulations, and adiabatic and cooled walls. Case 3 ($M = 2.5$, oblique $\beta = 0.1$) is a first mode; the rest are second (Mack) modes. Each anchor is Chebyshev-collocated at $N = 120$ –280 points (larger N for the thick $M = 10$ layers), domain height $y_{\max} = 40$ –140 δ^*/L^* , L^* scaling, with an isothermal disturbance-temperature condition and bulk-viscosity ratio $\lambda/\mu = 1.2$.

Both formulations of the collocation route appear, matched case by case to what Malik tabulates: the temporal cases 3–5 use the temporal GEVP (its 3-D variant for the oblique case 3), returning the complex frequency ω from a real wavenumber, while the spatial cases 6 and Table X use the spatial quadratic eigenvalue problem in companion form with shift–invert targeting, returning the complex wavenumber α from a real frequency. Table 3 lists the five anchors. Every real part (α_r or ω_r) is essentially exact, fixing each mode and its phase speed. The one outlier is Table X’s

spatial α_i , which reflects the documented hypersonic-second-mode inter-method spread: Tumin’s recomputation of case 6’s α_i from Malik’s printed value differs in the same way, and the solver of [8] gives -0.002780 . The *temporal* $M = 10$ growth rates (cases 4, 5) match closely, so the spatial α_i is the more formulation-sensitive quantity. Case 5 is Malik’s deliberately near-neutral “severe test”: his own SDSP scheme flips ω_i negative at coarse resolution, yet Our LST code lands on his converged value (Figure 2).

Table 3: The Malik [2] compressible eigenvalue anchors. Spatial cases (6, X) compare the complex wavenumber α ; temporal cases (3–5) compare the complex frequency ω .

Case	Conditions	Our LST code	Malik
6 (Tab. IX)	$M4.5$ adiab., spatial 2nd	$0.253400-0.002490 i$	$0.253405-0.002492 i$
3 (Tab. IV)	$M2.5$ oblique, temporal 1st	$0.036735+0.000584 i$	$0.036734+0.000584 i$
4 (Tab. V)	$M10$ cooled, temporal 2nd	$0.097457+0.001988 i$	$0.097484+0.002030 i$
5 (Tab. VI)	$M10$ adiab., temporal 2nd	$0.115862+0.000156 i$	$0.115865+0.000153 i$
X (Tab. X)	$M10$ cooled, spatial 2nd	$0.095803-0.002403 i$	$0.095933-0.002156 i$

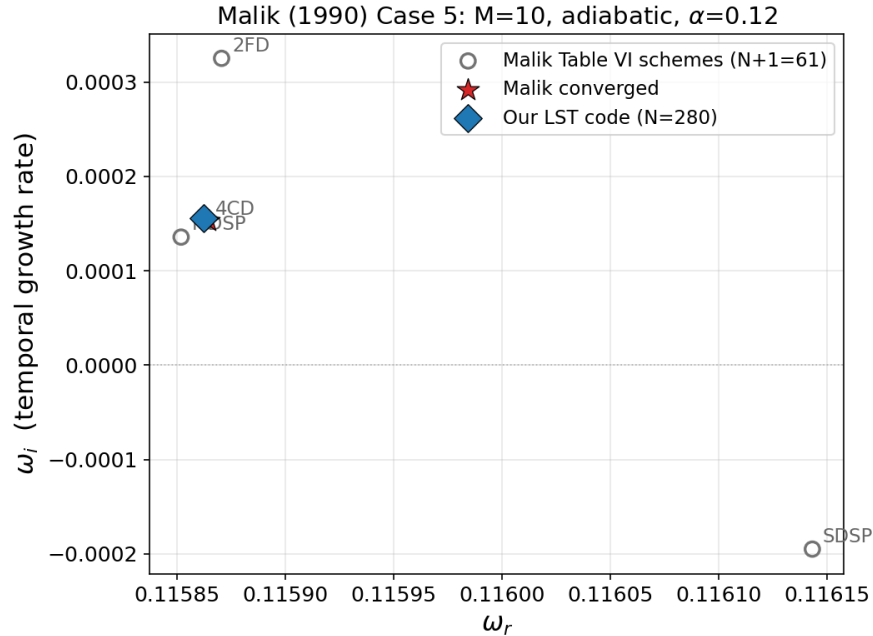


Figure 2: Malik case 5 [2] ($M = 10$ adiabatic, the deliberately near-neutral “severe test”): Our LST code (diamond) lands on Malik’s converged value (star) and his 4CD/MDSP cluster, while his own 2FD scheme sits high and SDSP flips ω_i negative; Our LST code reproduces the case Malik built to break solvers.

3.2 Malik (1990) Fig. 4: M10 second-mode eigenfunction

Reference. Malik [2], Fig. 4: the disturbance eigenfunctions of the same $M = 10$ cooled-wall condition as case 4. The comparison uses the temperature eigenfunction $\hat{T}$ (real and imaginary parts),

the dominant, cleanly traceable curves; the small velocity and pressure eigenfunctions overlap near the axis and are not reliably digitisable, so they are omitted.

At the case-4 conditions ($N = 200$ Chebyshev points, $y_{\max} = 75 \delta^*/L^*$, L^* scaling, isothermal disturbance-temperature condition) the temporal GEVP is read for its *eigenvector* rather than its eigenvalue: from a real wavenumber α it returns the discrete eigenpair, of which the temperature component $\hat{T}(y)$ is compared. This is therefore a mode-shape check, not an eigenvalue one, and the complex temperature eigenfunction $\hat{T}(y)$ is normalised so that $\hat{T}_r$ peaks at +1 (Malik's convention, which fixes the eigenvector's single global phase). Both the real and imaginary temperature eigenfunctions lie on the digitised Fig. 4 points (Figure 3). Our LST code's $\hat{T}_r$ traces the near-wall negative lobe, the zero crossing near $y \approx 6$, and the sharp unit peak at $y = 13.3$ ($\approx \delta^*$) against Malik's ≈ 13 ; $\hat{T}_i$ reproduces the same double-lobed shape, a positive hump near $y \approx 11$ and a negative dip near $y \approx 15$. Under the *single* global phase that makes $\hat{T}_r$ real at its peak, *both* parts fall on the reference, so the temperature eigenvector is validated in full, not merely its peak location. The temperature fluctuation dominates the velocity ($|\hat{T}|/|\hat{u}| \approx 14$), the hallmark of the hypersonic Mack mode.

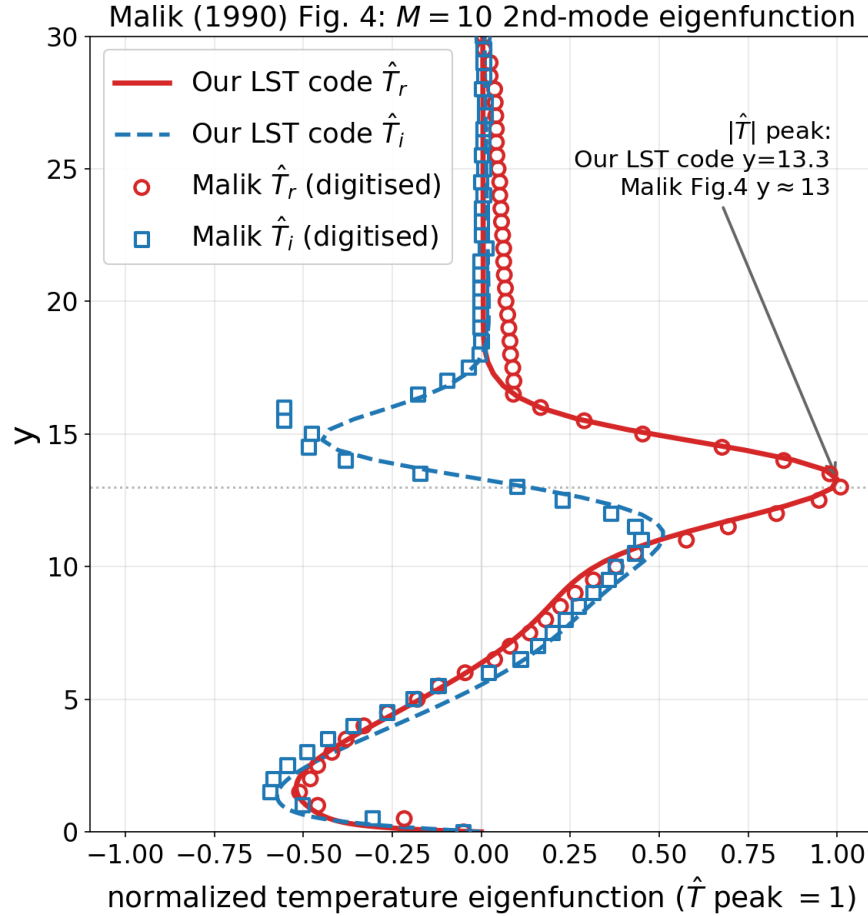


Figure 3: Malik Fig. 4 [2], $M = 10$ second-mode temperature eigenfunction: Our LST code's $\hat{T}_r$ (solid) and $\hat{T}_i$ (dashed) curves over the digitised points, normalised so $\hat{T}_r$ peaks at +1. Both parts match under a single global phase: $\hat{T}_r$ has its unit peak at $y = 13.3$ against Malik's ≈ 13 , and $\hat{T}_i$ shows the positive hump near $y \approx 11$ and the negative dip near $y \approx 15$.

3.3 Balakumar & Malik (1992): discrete/continuous spectrum topology

Reference. Balakumar & Malik [3]. The paper presents its branches graphically; no tabulated branch eigenvalues are openly recoverable, so this is a qualitative branch-topology check plus a fully specified spatial second-mode cross-check.

Conditions. $M = 4.5$, $T_0 = 311$ K, $Re = 1000$, $\omega = 0.2$, adiabatic (insulated) wall. The full companion spectrum is computed by Chebyshev collocation at $N = 120$ (121 nodes), $y_{\max} = 40 \delta^*/L^*$, L^* scaling, with an isothermal disturbance-temperature condition and $\lambda/\mu = 1.2$.

The spatial quadratic eigenvalue problem is solved for its *entire* companion spectrum, not merely the shift-invert neighbourhood: from a real frequency ω it returns all complex wavenumbers α of the discretised operator at once, discrete mode and rendered continuous spectrum together, which is what a branch-topology comparison requires. The comparison is thus twofold: the branch topology (how the discrete second mode separates from the continuous spectrum in the complex- α plane) and the discrete spatial eigenvalue α . Our LST code's full spatial spectrum reproduces the published topology: the discrete second mode sits at $\alpha = 0.220210 - 0.002787i$ ($c = 0.9082$), matching the published $0.220 - 0.003091i$, and is the unique least-damped root in the second-mode phase-speed band, while a cluster of roots accumulates at the freestream phase speed $c \rightarrow 1$ (the continuous spectrum). The discrete mode is set apart from that cluster in phase speed and growth rate, not by a wide gap in the complex plane: the discretised continuum is dense in α_r , so the nearest continuum root lies close by, the expected signature of the continuous spectrum crowding the discrete mode. Treated purely as a spatial second-mode anchor, the real part is essentially exact and the imaginary part carries the same inter-method α_i spread seen on Malik case 6 (Figure 4).

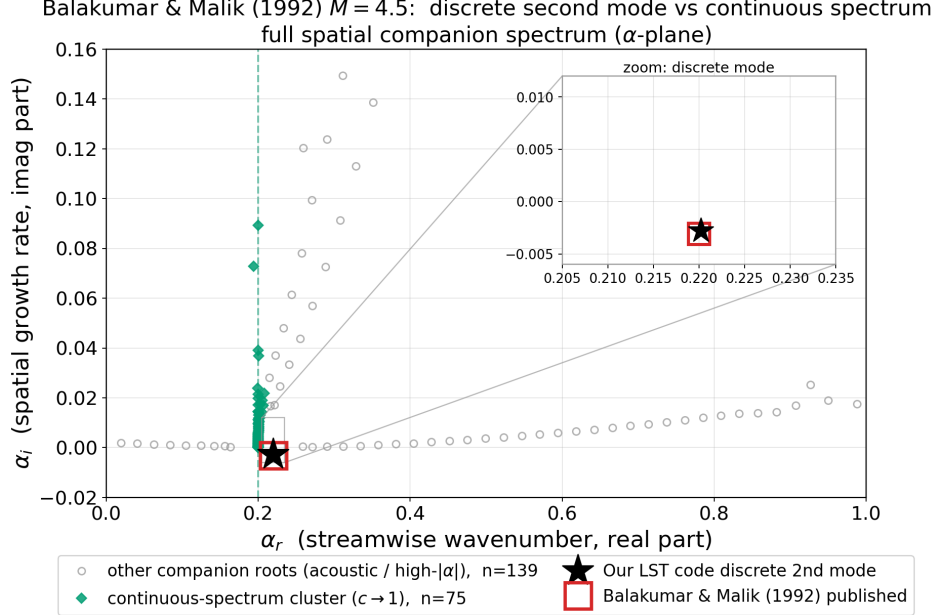


Figure 4: Balakumar & Malik [3], $M = 4.5$: Our LST code's complex- α spectrum showing the discrete second mode set apart (in phase speed and growth rate) from the continuous-spectrum cluster near $c \rightarrow 1$.

4 Second mode: curves and families

Having pinned the compressible operator at single points, the following cases test full growth-rate and neutral curves; the second mode is Our LST code’s design target.

4.1 Mack (1984) Fig. 10.6: second-mode growth vs. Reynolds number

Reference. Mack [4], Fig. 10.6.

Conditions. Air, adiabatic wall, 2-D wave ($\psi = 0^\circ$), at $M = 4.5, 5.8, 7$, and 10 . Two modelling choices make the comparison defensible: the cold edge state of Mack’s Table 11.1 [4] (resolving a historical $\sim 6\times$ gap that was an edge-temperature error) and a wall-normal domain $y_{\max} \approx 4\delta^*/L^*$ that grows with Mach ($y_{\max} = 40$ at M4.5 up to 140 at M10; a fixed short box starves the thick high-Mach boundary layer). The eigenvalue solves use Chebyshev collocation at $N = 110$, L^* scaling, and the 2-D Stokes closure $\lambda/\mu = 0$.

Sweeping the temporal GEVP in wavenumber at each Reynolds number (a real α scan at fixed R , from which the second-mode $\omega_i = \alpha c_i$ is maximised over α , then repeated over R) traces the curve. Across $M = 4.5, 5.8, 7$, and 10 , Our LST code’s maximum second-mode temporal growth rate $\omega_{i,\max}(R)$ (Mack’s L^* scale, $c_r \approx 0.9$) tracks the digitised Fig. 10.6 points. The match is tightest at $M = 7$ and $M = 10$ and carries a small residual at $M = 4.5$ and $M = 5.8$; because both ends of the Mach range match well, that residual is local rather than a monotonic high-Mach under-prediction. The $M = 4.5$ and $M = 10$ curves are shown as representatives (Figures 5 and 6); $M = 5.8$ and $M = 7$ follow the same trend and are summarised in Table 1.

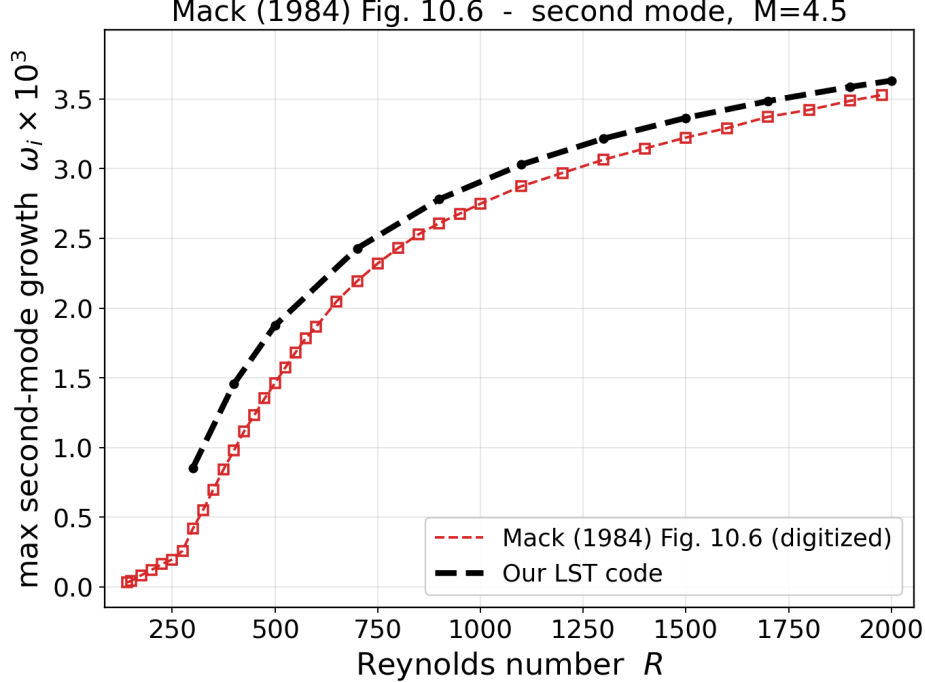


Figure 5: Mack Fig. 10.6 [4], $M = 4.5$: Our LST code’s $\omega_{i,\max}(R)$ over the digitised points, with a small residual at this Mach number.

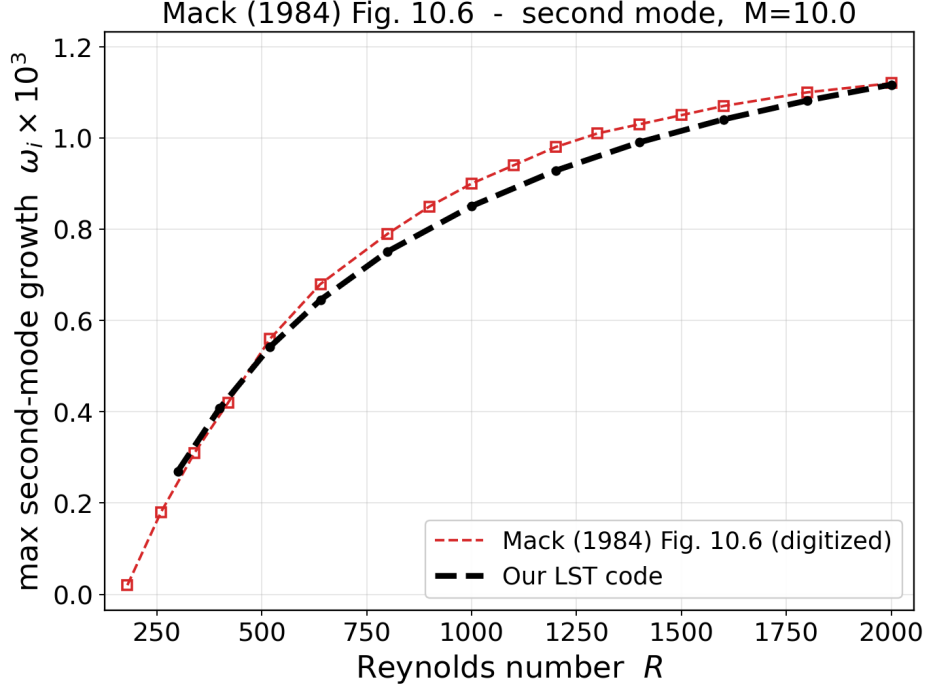


Figure 6: Mack Fig. 10.6 [4], $M = 10$: the same comparison at the top of the Mach range, with the domain-converged ($y_{\max} = 140$), cold-edge second mode.

4.2 Egorov et al. (2006): Mach-6 flat-plate second mode

Reference. Egorov, Fedorov & Soudakov [6]: 2-D DNS second-mode growth over a Mach-6 adiabatic flat plate, reported to agree with LST.

Conditions. Perfect gas $\gamma = 1.4$, $Pr = 0.72$, power-law viscosity $\mu/\mu_e = (T/T_e)^{0.7}$; $M_\infty = 6$, $Re_L = 2 \times 10^6$, adiabatic wall; mode-S circular frequency $\omega \approx 200$ (plate-length units) at the synchronisation location $x = 1$ ($R = \sqrt{2} \times 10^6 = 1414$). The spatial solve uses Chebyshev collocation at $N = 140$, $y_{\max} = 40 \delta^*/L^*$, L^* scaling, an isothermal disturbance-temperature condition, and $\lambda/\mu = 1.2$.

The spatial QEP is swept in frequency at the synchronisation location $R = 1414$ (real ω across the candidate band, returning the complex wavenumber α and hence the spatial growth $\sigma = -\alpha_i$ and its maximiser). Because the paper presents the DNS-vs-LST comparison as a growth-rate-vs- x curve with no printed growth-rate digit, the comparison is made on the second-mode instability band and the frequency of the most-amplified mode. Our LST code's discrete second mode ($c_r \approx 0.92$ – 0.94) is unstable across $\omega_E \in [140, 260]$, which brackets the reference forcing $\omega = 200$, and is most-amplified at $\omega_E \approx 215$; at exactly $\omega = 200$ the spatial growth is $-\alpha_i = 2.90 \times 10^{-3}$ with $c_r = 0.934$. Our LST code thus places the M6 second-mode instability squarely in the stated band, with the most-amplified frequency essentially on the reference forcing choice.

4.3 Ma & Zhong (2003) Fig. 15: M4.5 second-mode neutral curve

Reference. Ma & Zhong [7], §6 / Fig. 15: full neutral curves $\omega(R)$ for the first and second mode at $M = 4.5$, with the second-mode branch points quoted in text at $R = 806$ (branch I) and $R = 999.6$ (branch II) for $F = 2.2 \times 10^{-4}$.

Conditions. Air, adiabatic *mean* wall, 2-D ($\psi = 0^\circ$), $M = 4.5$, $T_e = 65.15$ K, unit $Re = 7.2 \times 10^6/\text{m}$, Sutherland $S = 110.33$ K, $Pr = 0.72$, $\gamma = 1.4$; spatial formulation at fixed $F = 2.2 \times 10^{-4}$. The *disturbance* temperature wall condition is isothermal ($\hat{T}|_w = 0$), which the authors state they used for the Fig. 15 neutral curve. The spatial sweeps use Chebyshev collocation at $N = 130$ (131 nodes), $y_{\max} = 40 \delta^*/L^*$, L^* scaling, and $\lambda/\mu = 1.2$.

The spatial QEP is run in fixed-frequency sweeps: real ω along each $F = \text{const}$ path in R returns the complex wavenumber, and the neutral curve is the $\sigma = -\alpha_i = 0$ crossing set; the first-mode branches use the same route with the discrete-mode selector (freestream decay + $y_{\max}$ -stationarity). Our LST code thus traces the full neutral curve $\omega(R)$, anchored against the two second-mode branch points quoted in the text at $F = 2.2 \times 10^{-4}$. Its second-mode neutral band reproduces the closed loop of Fig. 15 (nose near $R \approx 280$, upper branch $\omega \approx 0.225$, lower branch $\omega \approx 0.18$) across the full range. At the quoted $F = 2.2 \times 10^{-4}$, the band crosses at $R = 786.6$ (branch I; text value 806) and $R = 1009.9$ (branch II; text value 999.6). Using the adiabatic disturbance condition by mistake shifts these to 831 and 1033; the paper’s actual isothermal disturbance condition is a correction, not a fit, and tightens the match. The first-mode *cutoff* (upper) branch, re-extracted with the discrete-mode selector (freestream decay + $y_{\max}$ -stationarity, replacing the earlier phase-speed-band classifier), follows Fig. 15’s upper branch closely. The low- ω *onset* (lower) branch is *not* recovered: below $\omega \approx 0.03$ the 2-D first mode ($c_r \lesssim 0.45$) delocalises into the slow-acoustic continuous spectrum ($c_r \leq 1 - 1/M = 0.778$), leaving no clean discrete eigenmode. This limit is the same across Our LST code’s spatial and temporal collocation solvers and a QR-shooting variant, and a wave-angle sweep confirms the 2-D mode is the most unstable here (so this is a 2-D, not an oblique, first mode). Recovering the onset branch would need a compound-matrix shooting solver robust to the near-continuous-spectrum decay rates, which is left as an open item in the companion working note (Figure 7).

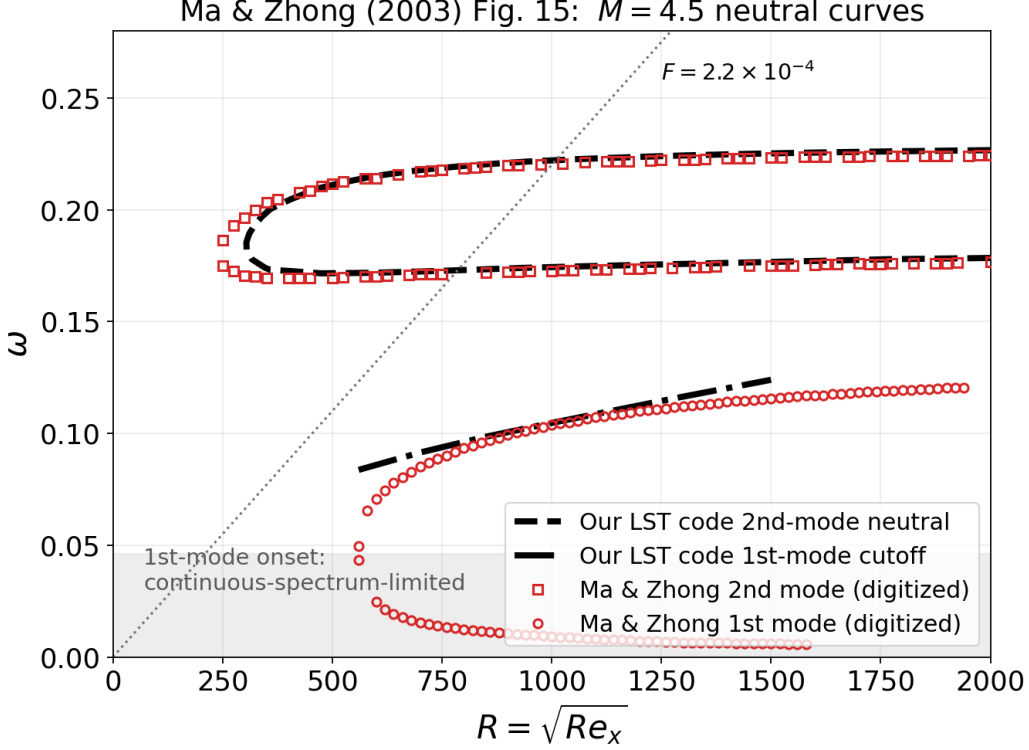


Figure 7: Ma & Zhong Fig. 15 [7], $M = 4.5$: Our LST code’s second-mode neutral loop over the pixel-digitised Fig. 15 points, matching the branch points at $F = 2.2 \times 10^{-4}$ ($R = 786.6$ vs. 806; $R = 1009.9$ vs. 999.6) under the isothermal disturbance condition. The first-mode *cutoff* branch, re-extracted with the discrete-mode selector, follows the digitised upper branch; the first-mode *onset* (shaded band) is continuous-spectrum-limited, the long-wavelength mode delocalising into the continuous spectrum, a limit shared by all of Our LST code’s collocation and QR-shooting methods (see text).

5 First mode: curves and families

The first mode was Our LST code’s documented weak spot. After a reference-digitisation correction, most of the Mack first-mode figures now match well; the remaining disagreement is narrow and isolated to the $M \approx 2.2$ cases (Fig. 10.1 and Fig. 10.3 at $M = 2.2$), a genuine under-amplification and too-high critical Reynolds number at that Mach rather than a general trend, and is tracked in the companion working note. This section reports the Mack first-mode figures; the Özgen–Kırcah neutral-curve family, whose figures carry first-mode branches alongside second-mode ones, is treated separately in Section 6.

5.1 Mack (1984) Fig. 10.1: first-mode neutral loop

Reference. Mack [4], Fig. 10.1, the innermost (complete-equations) neutral loop, re-digitised with both branches plus nose.

Conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$), at $M = 1.6$ and $M = 2.2$; temporal neutral loop ($c_i = 0$) in $(R, F \times 10^4)$. The (R, α) -grid solves use Chebyshev collocation at $N = 140$, $y_{\max} = 45 \delta^*/L^*$, L^* scaling, and $\lambda/\mu = 0$.

Solving the temporal GEVP over an (R, α) grid (real α at each R , returning the complex phase speed) gives the neutral loop as the $c_i = 0$ contour, with the critical Reynolds number at its nose. At $M = 1.6$ the first-mode neutral loop (its lower onset and upper cutoff branches) closely tracks Mack’s complete-equations loop after the corrected reference, with critical Reynolds number $R \approx 300$ against Mack’s ≈ 215 (Figure 8). The companion $M = 2.2$ loop shows a genuine first-mode under-amplification, with a too-high critical Reynolds number ($R \approx 400$ against Mack’s ≈ 300), and is documented in the companion working note.

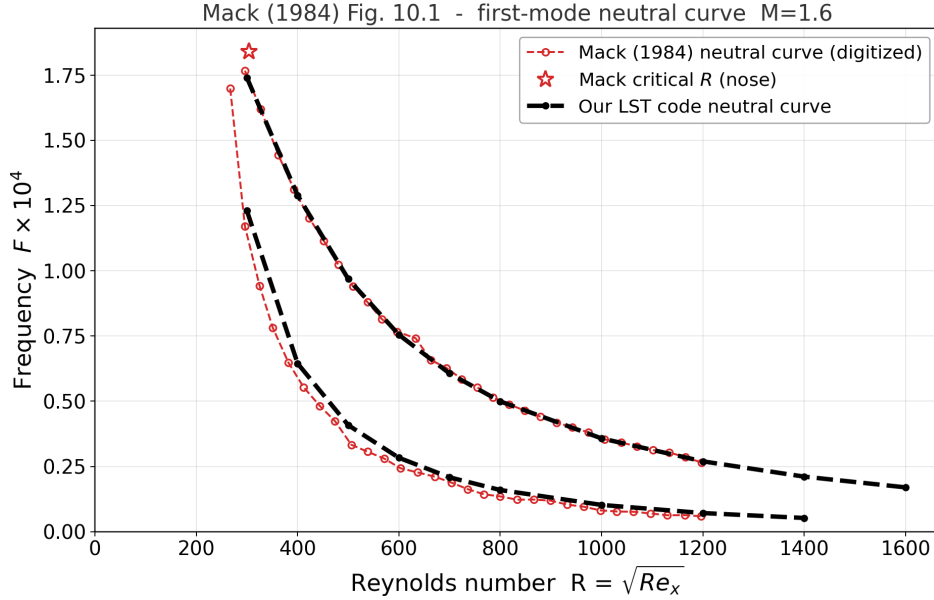


Figure 8: Mack Fig. 10.1 [4], $M = 1.6$: Our LST code’s first-mode neutral loop closely tracks Mack’s complete-equations loop, with critical Reynolds number $R \approx 300$ against Mack’s ≈ 215 .

5.2 Mack (1984) Fig. 10.3: oblique first-mode growth vs. R

Reference. Mack [4], Fig. 10.3.

Conditions. Air (ideal, $Pr = 0.72$), adiabatic wall; oblique waves at $M = 1.3$ ($\psi = 45^\circ$), $M = 2.2$ ($\psi = 45^\circ$), $M = 3.0$ ($\psi = 60^\circ$). These oblique cases use the exact 8×8 shooting march rather than a Chebyshev grid, with $y_{\max} \approx 4 \delta^*/L^*$ and L^* scaling.

These curves use the exact oblique first-order *shooting* route rather than the collocation GEVP: at these Mach numbers the reduced temporal EVP’s branch selection is not trusted, so an anchor root is located by the $\sigma_{\min}(c) \rightarrow 0$ search and continued in the parameters, with the growth maximised over wavenumber at each R . At $M = 1.3$ Our LST code’s maximum first-mode temporal growth $\omega_{i,\max}(R)$ tracks Fig. 10.3 closely and reproduces the rigorous Table-10.1 eighth-order anchor; $M = 3.0$ tracks it as well. $M = 2.2$ is the exception, a genuine under-amplification (convergence-checked: not a $y_{\max}$ or step-count artifact). Topology is correct in all three (a single rising-then-turning first-mode branch), so $M = 2.2$ is an isolated outlier rather than part of a growing-with-Mach trend. The $M = 1.3$ case is shown in Figure 9; the $M = 2.2$ case is documented in the companion working note.

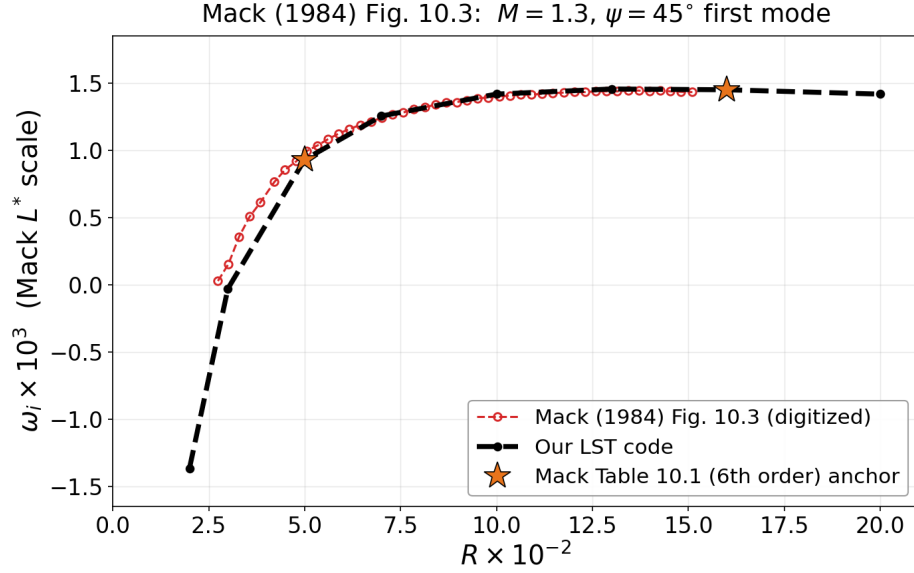


Figure 9: Mack Fig. 10.3 [4], $M = 1.3$, $\psi = 45^\circ$: Our LST code’s $\omega_{i,\max}(R)$ over the digitised points, reproducing the rigorous Table-10.1 eighth-order anchor.

5.3 Mack (1984) Fig. 10.4: oblique first mode at high Mach

Reference. Mack [4], Fig. 10.4.

Conditions. Air, adiabatic wall; the high-Mach first mode is oblique-dominated, so growth is maximised over both α and wave angle ψ ; $M = 4.5, 5.8, 7, 10$ with $y_{\max} \approx 4\delta^*/L^*$. The 3-D temporal solves use Chebyshev collocation at $N = 200$ ($y_{\max}$ reaching $150\delta^*/L^*$ at $M = 10$), L^* scaling, and $\lambda/\mu = 0$.

The reduced 3-D temporal GEVP takes the real oblique pair α and wave angle ψ (equivalently β) and returns the complex frequency, with ω_i maximised over (α, ψ) at each R . Across $M = 4.5, 5.8, 7$, and 10 , Our LST code’s maximum first-mode temporal growth $\omega_{i,\max}(R)$ tracks the digitised Fig. 10.4 points, with the oblique-first-mode peak near $\psi \approx 60^\circ$ ($c_r \approx 0.83$ at $M = 4.5$) and only a small residual at $M = 10$, the top of the Mach range. One caveat: these curves were computed with the reduced dense-EVP path rather than the exact-shooting path used for the convergence-checked Fig. 10.3 cases, and no exact-shooting cross-check was recorded; the domain used the second-mode-calibrated $\sim 4\delta^*/L^*$ rule, while the first mode reaches further from the wall. There is no first-mode under-amplification trend at high Mach in this figure. $M = 4.5$ and $M = 10$ are shown (Figures 10 and 11); $M = 5.8$ and $M = 7$ follow the same trend and are summarised in Table 1.

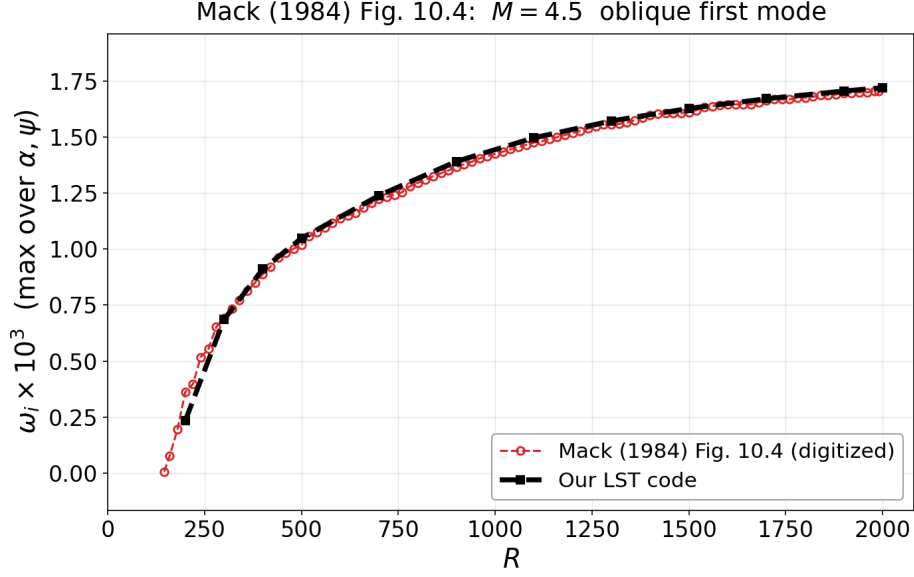


Figure 10: Mack Fig. 10.4 [4], $M = 4.5$: the oblique first mode optimised over (α, ψ) , over the digitised points.

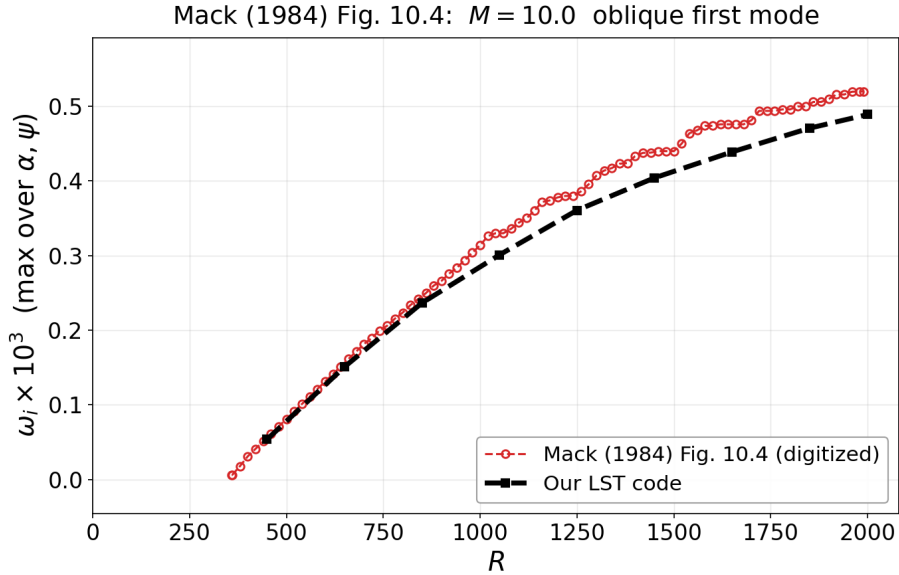


Figure 11: Mack Fig. 10.4 [4], $M = 10$: the same comparison at the top of the Mach range, with a small residual (lower-confidence magnitude, see text).

6 Mixed first and second mode: the Özgen–Kırcalı neutral-curve family

Fig. 3 of Özgen & Kırcalı [5] presents, for each Mach number from 2 to 10, a single neutral-curve figure in the (Re_L, α_L) plane. At $M = 2$ and 3 the figure contains the first-mode lobe alone; from $M \approx 4$ the second (Mack) mode enters, and each figure then spans *both* instability families; at

$M = 6$, for example, four branches: second-mode upper and lower, first-mode cutoff, and first-mode onset. Because a single reproduced figure tests both modes at once, these cases are classified neither under the first nor the second mode but as a mixed family, and they are reported last, drawing on both preceding sections: at $M = 2$ and $M = 3$ only the first mode is present, while $M = 4$ – 10 span both modes.

Reference. Özgen & Kırçalı [5], Fig. 3 ($M = 2, 3, 4, 6, 7, 8, 10$).

Conditions. Air, adiabatic wall, 2-D ($\psi = 0^\circ$); temporal neutral contour ($c_i = 0$) in the (Re_L, α_L) plane, with the Özgen temperature-dependent transport laws. The neutral loci are traced on two Chebyshev grids: the wall-trapped second mode at $N = 180$ with $y_{\max} = 8$ – $12 \delta^*/L^*$, and the more diffuse first mode at $N = 200$ with $y_{\max} = 35$ – $45 \delta^*/L^*$; L^* scaling throughout.

The temperature-form 2-D temporal GEVP (the Özgen equation arrangement) is solved over a (Re_L, α_L) grid, real α_L at each Re_L returning the complex phase speeds. A discrete-mode extractor (eigenfunction decay + $y_{\max}$ -stationarity, the two acceptance tests of the companion numerical-methods document) selects the physical mode at each node, eigenvalue continuation carries difficult branches through the grid, and the neutral family (all branches $\alpha_L(Re_L)$ of both the first and second mode) is the extracted $c_i = 0$ contour. This extractor replaces the earlier c_r -band classifier that had excluded the genuine first mode; with that correction and the references re-digitised, the picture is:

- $M = 4$ **and** $M = 6$ (Figures 12 and 14): a complete two-lobe match. At $M = 4$ both second-mode branches sit on the digitisation floor and the first-mode cutoff is reproduced; only the deep low- α onset departs. At $M = 6$, after re-digitising the reference and re-tracing the neutral loci on a grid extended to low Re , Our LST code closes both lobes at their critical Reynolds number (second-mode nose $R \approx 206$ against Özgen's ≈ 200) and reproduces the three robust branches (second-mode upper and lower, first-mode cutoff); only the first-mode onset runs below the reference, the continuous-spectrum-limited tail.
- $M = 2$ **and** $M = 3$ (Figure 13): $M = 2$ traces both first-mode branches over the full R -range by eigenvalue continuation; at $M = 3$ the first-mode cutoff is reproduced where resolvable, the onset again being continuous-spectrum-blocked.
- $M = 7$, $M = 8$, $M = 10$: the dominant second-mode cutoff branch is reproduced, but Our LST code predicts a *connected* first-second-mode band where the reference shows two lobes separated by a stable gap. This is a connected-vs-separated topology difference (inter-mode over-prediction at high Mach), not a second-mode-curve error.

A separate growth-contour comparison (the $c_i = 0.004$ lobe rather than the neutral curve) does not match, and the interior $c_i = 0.012$ M2 lobe is absent because Our LST code's peak M2 growth $c_{i,\max} = 0.0109$ falls just short of 0.012; this is the same open-lobe / under-amplification discrepancy noted above. Representative cases for $M = 2, 4$, and 6 follow (Figures 13, 12, and 14); the remaining Mach numbers ($M = 3, 7, 8$, and 10) are summarised in Table 1.

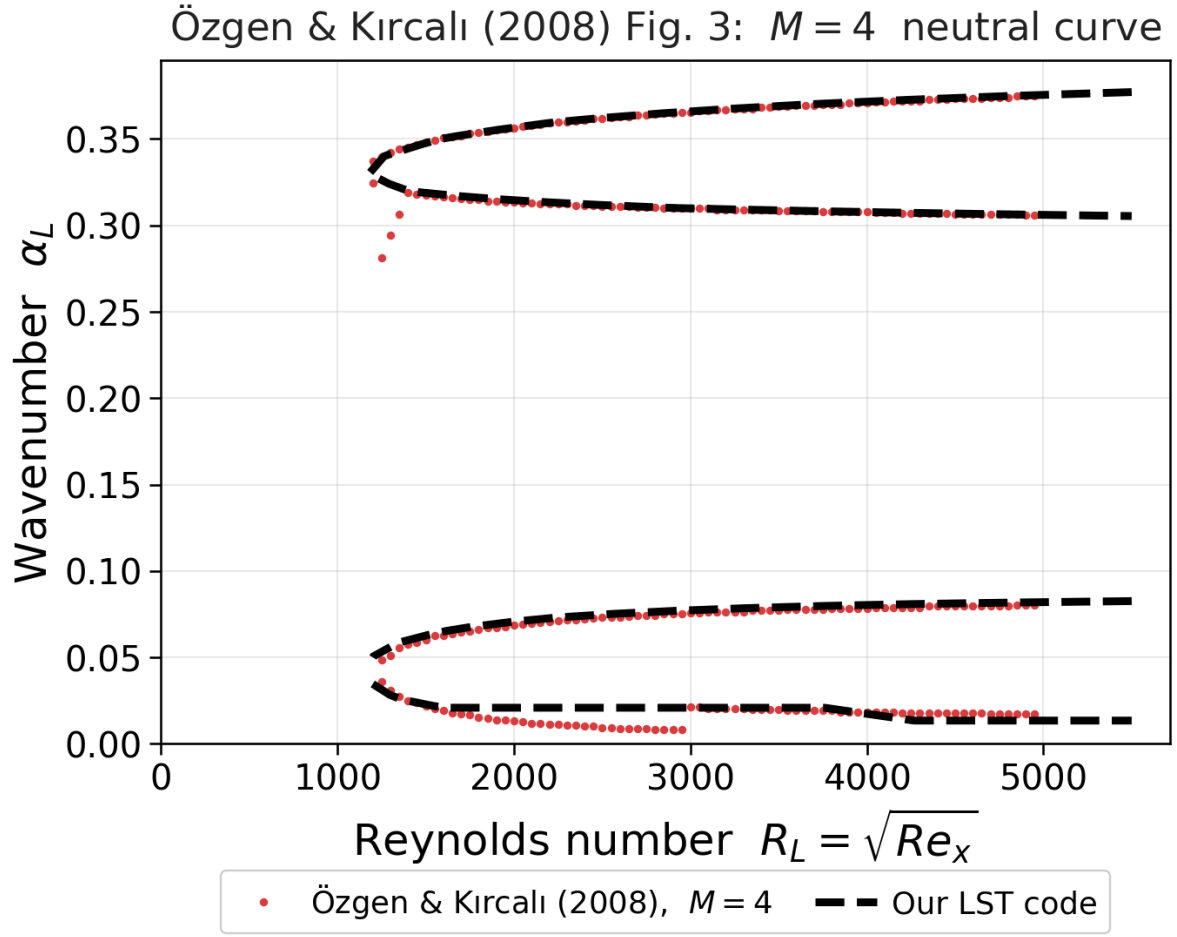


Figure 12: Özgen & Kırçalı Fig. 3 [5], $M = 4$: Our LST code reproduces the entire two-lobe neutral structure, with both second-mode branches on the digitisation floor and the first-mode cutoff closely tracked.

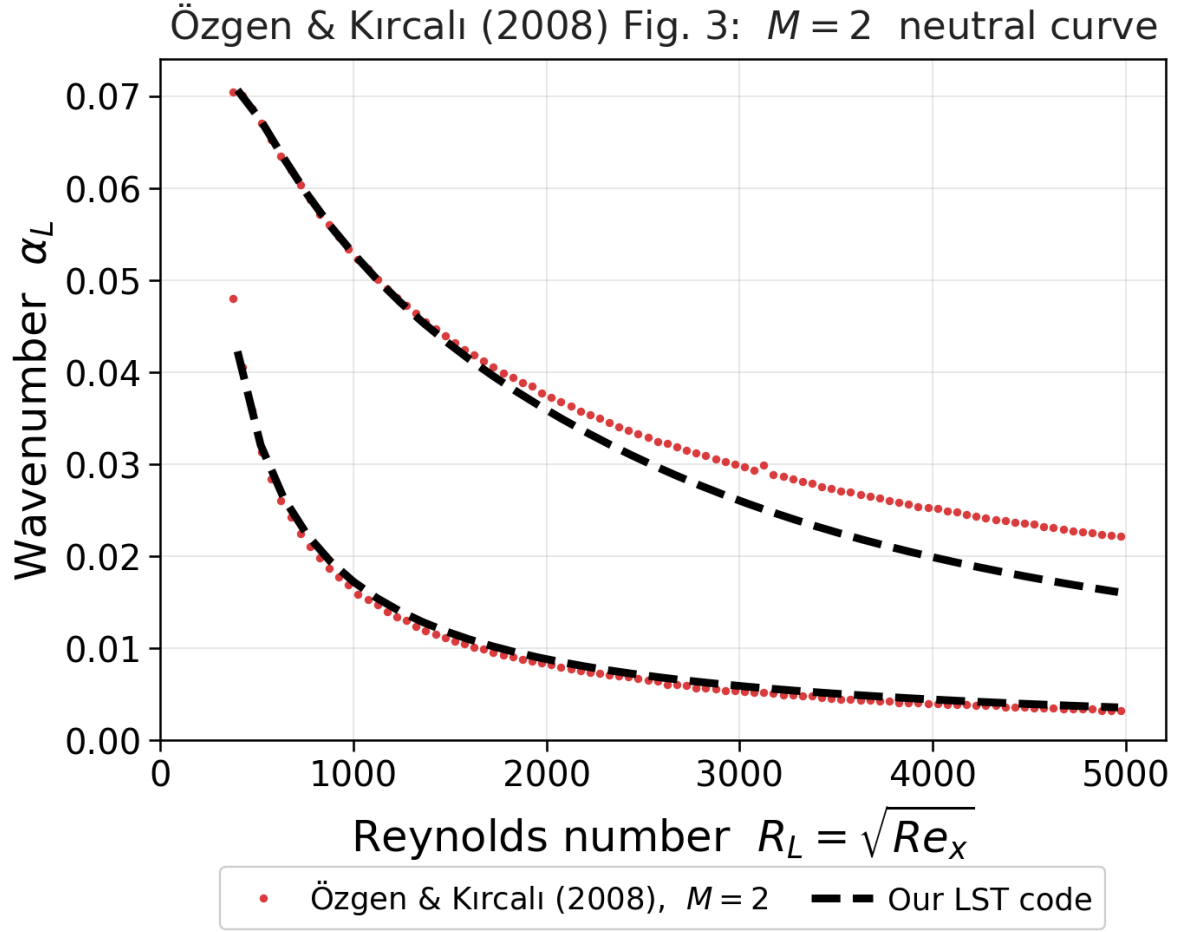


Figure 13: Özgen & Kırçalı Fig. 3 [5], $M = 2$ (first mode only at this Mach): both branches traced over the full R -range by eigenvalue continuation.

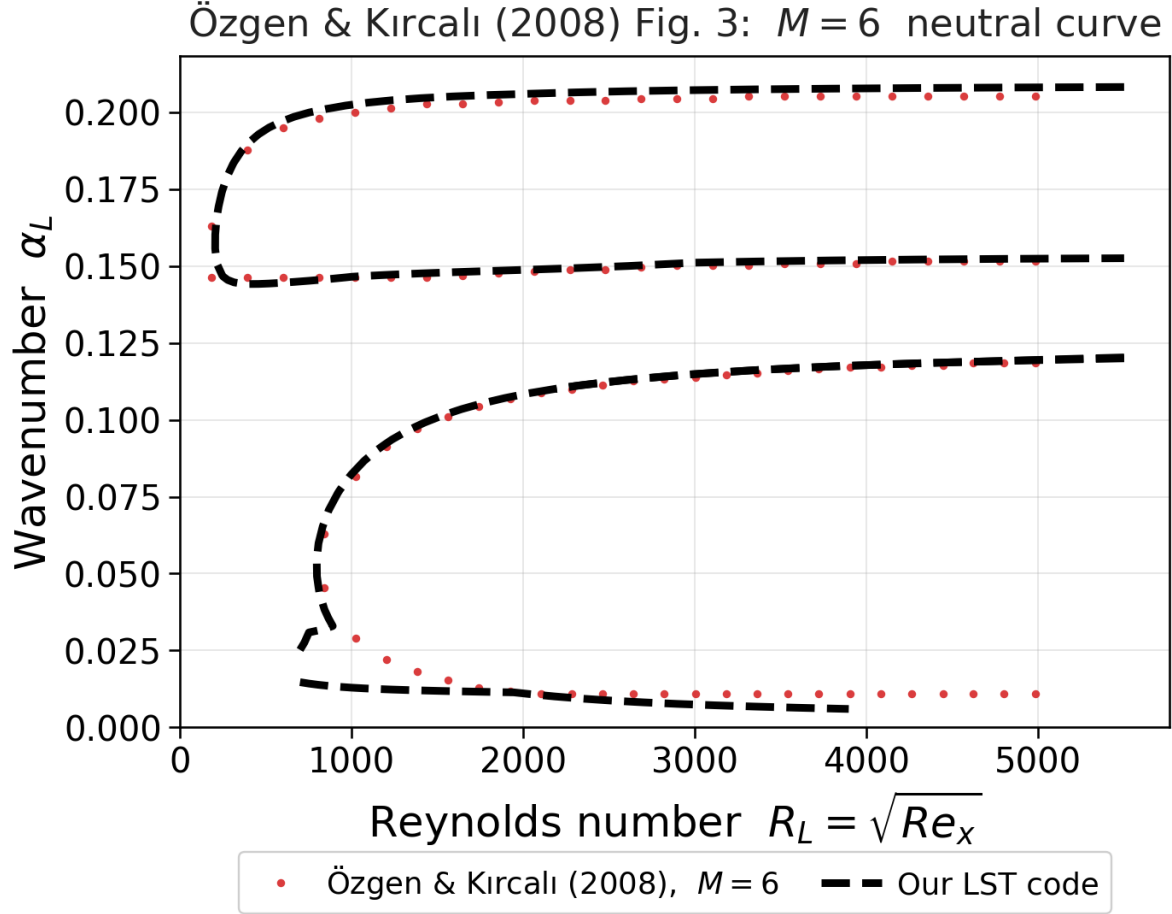


Figure 14: Özgen & Kırçalı Fig. 3 [5], $M = 6$ (both modes in one figure): the re-digitised reference points and Our LST code’s full $c_i = 0$ contour. Re-tracing the neutral loci on a grid extended to low Re closes both lobes at their critical Reynolds number, the second-mode nose sitting at $R \approx 206$ against Özgen’s ≈ 200 . The three robust branches (second-mode upper and lower, first-mode cutoff) are reproduced; only the deep low- α first-mode onset runs below the reference, the continuous-spectrum-limited tail.

Cases that are not settled validations (the isolated $M \approx 2.2$ first-mode disagreements, the sharp-cone N -factor with its bounded magnitude deficit, the private Mach 5.35 code-to-code benchmark, and the Mach 6 pipeline demonstration) are collected in the companion working note.

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